Scaling Spherical CNNs

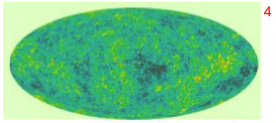
Carlos Esteves Jean-Jacques Slotine Ameesh Makadia
machc@google.com jjs@mit.edu makadia@google.com

github.com/google-research/spherical-cnn

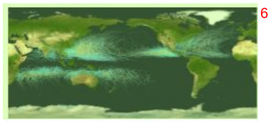
Introduction [1]

Spherical CNNs [2]

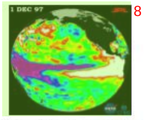
- Principled way of learning from spherical signals, using spherical convolutions computed spectrally. [3]
- Equivariant to 3D rotations.
- Respects spherical topology, robust to sampling.



Cosmic radiation (NASA / WMAP Science Team) [4] [5]



Cyclone tracks (NASA, Nilfanion) [6] [7]



Sea surface temperature [8] [9]

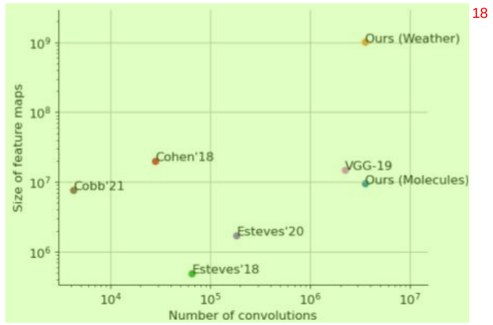
Spherical panorama (SUN360) [10] [11]

Molecules [12] [13]

3D CAD models (Modelnet40) [14] [15]

Contributions [16]

- Spherical CNNs 100x larger than previously possible. [17]
- More expressive and efficient layers and activations.
- Application-specific modeling for molecules and weather.



[18]

Related work [19]

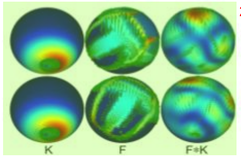
Cohen et al (ICLR'18), Esteves et al (ECCV'18, NeurIPS'20), Kondor et al (NeurIPS'18), [20]
Cobb et al (ICLR'21), McEwen et al (ICLR'22), Ocampo et al (ICLR'23). [21]

Method [22]

- Based on spin-weighted SCNNs (Esteves et al, NeurIPS'20). [23]
- Phase collapse activation: $x_0 \leftarrow W_1 x_0 + W_2 |x| + b$
  - Inspired by Guth et al, arXiv'21. [24]
- Efficient spin-weighted spherical harmonics on TPUs [25]
  - Avoid tensor slicing/concatenation.
  - Use DFT matrices instead of FFT.
- Spectral batch normalization, pooling and residual connection.

[26]

Spectral residual block [27]



[28]

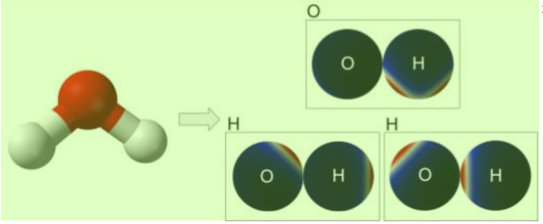
Spin-weighted spherical convolution with spin=1 as a vector field [29]

[30]

	Δ Steps/s [%] ↑	Δ RMSE [%] ↓
JAX implementation	33.7	0.0
Phase collapse	−4.6	−8.0
No Δ symmetries	16.3	0.0
Use DFT	21.4	0.0
Spectral batch norm	7.8	−1.4
Efficient residual	19.3	−2.4

Molecular property regression [31]

- Represent a molecule as a set of spheres per atom, regress a set of properties. [32]
- One feature channel per atom type.
- Use directional attributes only (no radius).
- $f_{iz}(x) = \sum_{z_j = z} \frac{z_i z_j}{|r_{ij}|^p} e^{-\frac{(x \cdot r_{ij})^2}{\beta |r_{ij}|}}$ [33]



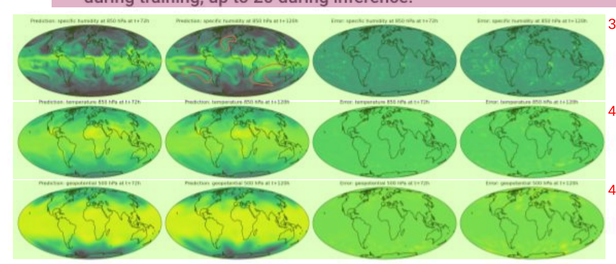
[34]

[35]

		μ [D]	α [a_0^3]	ϵ_{HOMO} [meV]	ϵ_{LUMO} [meV]	ϵ_{gap} [meV]	$< R^2 >$ [a_0^2]	zpve [meV]	U_0 [meV]	U [meV]	H [meV]	G [meV]	C_v [$\frac{\text{cal}}{\text{mol K}}$]
Split 1	DimeNet++ (2020)	0.030	0.044	24.6	19.5	32.6	0.331	1.21	6.32	6.28	6.53	7.56	0.023
	PaiNN (2021)	0.012	0.045	27.6	20.4	45.7	0.066	1.28	5.85	5.83	5.98	7.35	0.024
	TorchMD-Net (2022)	0.011	0.059	20.3	17.5	36.1	0.033	1.84	6.15	6.38	6.16	7.62	0.026
	Ours	0.016	0.049	21.6	18.0	28.8	0.027	1.15	5.65	5.72	5.69	6.54	0.022
Split 2	EGNN (2021)	0.029	0.071	29.0	25.0	48.0	0.106	1.55	11.00	12.00	12.00	12.00	0.031
	SEGNN (2022)	0.023	0.060	24.0	21.0	42.0	0.660	1.62	15.00	13.00	16.00	15.00	0.031
	Equiformer (2022)	0.014	0.056	17.0	16.0	33.0	0.227	1.32	10.00	11.00	10.00	10.00	0.025
	Ours	0.017	0.049	22.3	19.1	29.8	0.028	1.19	5.96	5.98	5.97	6.97	0.023

Weather forecasting [36]

- Take a set of atmospheric variables like temperature, wind, humidity, and predict their future values with fully convolutional models. [37]
- WeatherBench (Rasp et al): predict directly 3 or 5 days ahead.
- Keisler (arXiv'22): predict iteratively in 6h increments.
  - Our largest model: 256x256x78 inputs and outputs; iterated for 12 steps during training, up to 20 during inference. [38]



[39] [40] [41]

[42]

	1 day		3 days		5 days	
	Z500	T850	Z500	T850	Z500	T850
Keisler (2022)	64.9	0.730	175.5	1.17	344.7	1.78
Ours	58.3	0.827	167.2	1.26	340.0	1.91

[43]

	3 days			5 days		
	Z500 [m^2/s^2]	T850 [K]	T2M [K]	Z500 [m^2/s^2]	T850 [K]	T2M [K]
2 predictors						
Rasp et al. (2020)	626	2.87	-	757	3.37	-
Ours	531	2.38	-	717	3.03	-
117 predictors						
Rasp & Thuerey^{cont}	331	1.87	1.60	545	2.57	2.06
Rasp & Thuerey	314	1.79	1.53	561	2.82	2.32
Ours	329	1.62	1.29	601	2.57	1.89

[44] Similar patterns may appear at different positions and orientations; equivariance allows filter reuse. [45]